a) $\mathrm{N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3}$

b) $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

c) $\int_0^{\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$

d) $P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$

e) $F = G\frac{m_1 m_2}{r^2}$

f) $i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t)\right]\Psi(\mathbf{r}, t)$

g) $\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$

h) $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

i) $\mathrm{Fe^{2+} \longrightarrow Fe^{3+} + e^-}; \qquad \mathrm{MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O}$

j) $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

k) $H(X) = -\sum_{i=1}^{n} p(x_i) \log_2 p(x_i)$